Rabin Karp

97-ASCII

a - 1	e - 5	i - 9
b - 2	f - 6	j - 10
c - 3	g - 7	
d - 4	h - 8	

Text $n=6$ a a a a a b

P $m=3$ a a b

$\underbrace{1+1+2}_{\text{hash fn}} = 4$ hash code

h(P)

a a a a a b

$1+1+1=3$

Slide

$1+1+1=3$

Rolling hash fun

$1+1+1=3$

$1+1+2=4$

Eg T= a b c d a b c e

P= bce

$2+3+5 = 10$

$1+2+3=6$

$6-1=5+4=9$

$9-2+1=8$

$8-3+2=7$

$7-4+3=6$

$6-1+5=10$

$O(n-m+1)$

Drawback

E.g. c c a c c a a e d b a

P=3

dba

$4+2+1=7$

$3+3+1=7$ no match

$3+1+3=7$ "

$1+3+3=7$ "

$3+1+1=5$

$1+1+5=7$ no match

"Spurious Hits"

$O(mn)$

Eg 3

i++, j++

a a a a a a b

$\quad\quad$ 1 2 3 4 5 6 7 8 9 10 11

Eg3 c c a c c a a e d b a

P : d b a

m=3

$4\times10^{2} + 2\times10^{1} + 1\times10^{0}$

Robin finger Print fun.

$P[1]\times10^{m-1} + P[2]\times10^{m-2} + P[3]\times10^{m-3}$ ← ∵ we consider only 10 alphabet all alphabets.

$400+20+1 = 421$

c c a

$3\times10^{2} + 3\times10^{1} + 1\times10^{0} = 331$

cac

$\left[\left[3\times10^{2}+3\times10^{1}+P\times10^{0}\right] - 3\times10^{2}\right]\times10 + 3\times10^{0}$

$[331-300]\times10 + 3\times10^{0}$

$310+3 = 313$

acc

$[313-300]\times10 + 3\times10^{0}$

$130+3 = 133$

cca

$[133-100]\times10 + 1\times10^{0} =$

$330+1 = 331$

caa

$[331-300]\times10 + 1\times10^{0} =$

$310+1 = 311$

aae

$[311-300]\times10 + 5\times10^{0}$

$110+5 = 115$

aed

$[115-100]\times10 + 4\times10^{0}$

$150+4 = 154$

edb

$[154-100]\times10 + 2\times10^{0}$

$540+2 = 542$

dba

$[542-500]\times10 + 1\times10^{0}$

$420+1 = \underline{421}$

If we thing the fun value exceeds the capacity of the variable ~~sizeof the data bytes~~ we may use % fun.

$[\quad\quad] \% 2^{32}$

for 32 bit

But there is a chance of Spurious Hits

$O(n-m+1)$

Worst case

$O(mn)$